

50.007 Machine Learning

Support Vector Machine I

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Self-Introduction

- Call me Roy (Please avoid calling me Dr. Roy)
- Research: [Social AI Studio](#)
- Contact:
 - roy_lee@sutd.edu.sg
 - @sroylee (telegram)

User Profiling
in Multiple
Social Media

Online
Misbehavior &
Disinformation
Mining

Social Natural
Language
Generation

Social
Recommender
Systems

Happenings in Rest of the Course

Week	Topic	Instructor
4	Support Vector Machines (I) Support Vector Machines (II)	Roy – HW2
5	Logistic Regression Neural Networks	Roy
6	Gaussian Mixture Models Midterm Naive Bayes + Project Briefing	Roy
7	Recess Week	
8	Decision Tree and Random Forest Ensemble Methods	Roy
9	Dimensionality Reduction – PCA (I) Dimensionality Reduction – Autoencoder (II)	Roy
10	Hidden Markov Models (I) Hidden Markov Models (II)	Roy
11	Markov Decision Process (TBC – may swap with week 12) Reinforcement Learning (TBC – may swap with week 12)	Malika
12	Guest Lecture (Wed, 31 July, 12:30 - 1:30pm) Project Consultation	DSO/Roy
13	Guest Lecture (Wed, 07 Aug, 12:30 - 1:30pm) Project Presentations	Microsoft/Roy

Tips to do well in course

1. **Be curious** - Ask questions in class!
2. **Be adventurous** - Venture beyond class material for your project!
Google/YouTube for interesting materials!
3. **Be practical** - Think about the concepts and write codes to test it out!
4. **Be collaborative** - Discuss your ideas and doubts with your peers and me!
 - a. I have a lot of respect for students who share and help peers.
 - b. PS: I value non-anonymous engagement because I usually remember the student who asked me questions or illuminated my mind (easier to write recommendation letter).

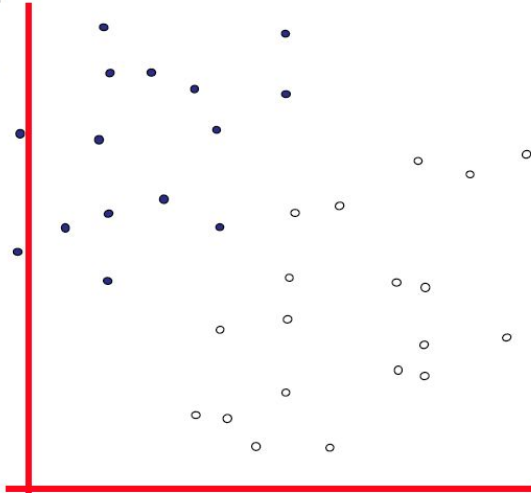
Big Picture of SVM

Support Vector Machines

- Suppose you have a linear classifier

Class labels

- denotes +1
- denotes -1



How will you draw
the decision
boundary?

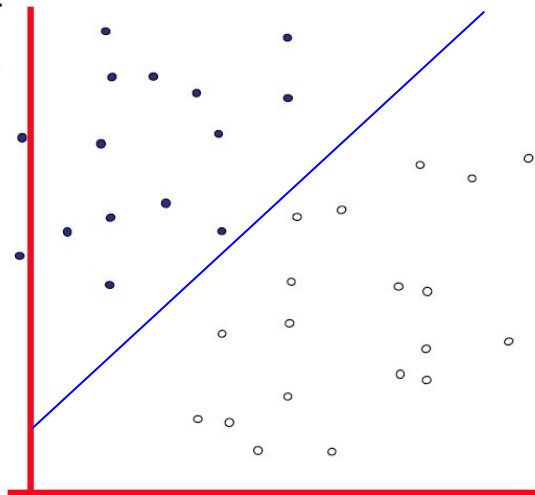


Support Vector Machines

- Suppose you have a linear classifier

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You might do this...

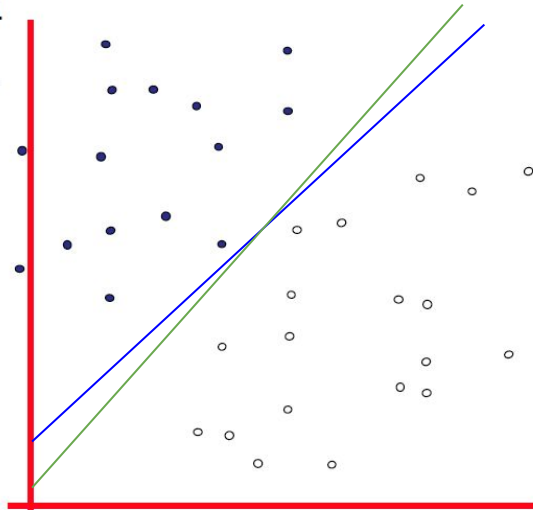


Support Vector Machines

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Or this...

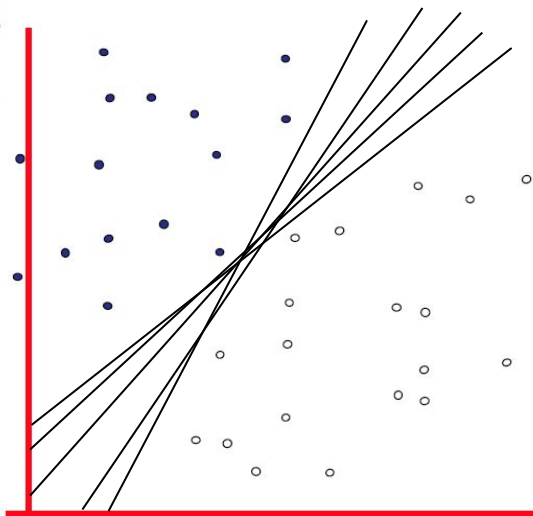


Support Vector Machines

- Suppose you have a linear classifier

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In fact all these are possible too...

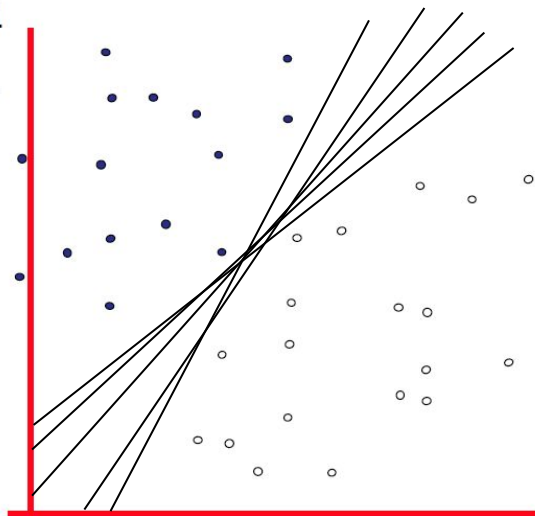


Support Vector Machines

- Suppose you have a linear classifier

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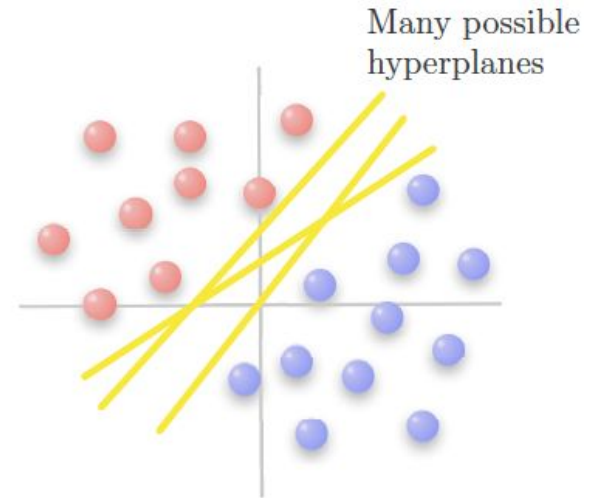


So, given these possibilities, **which decision boundary is the best?**



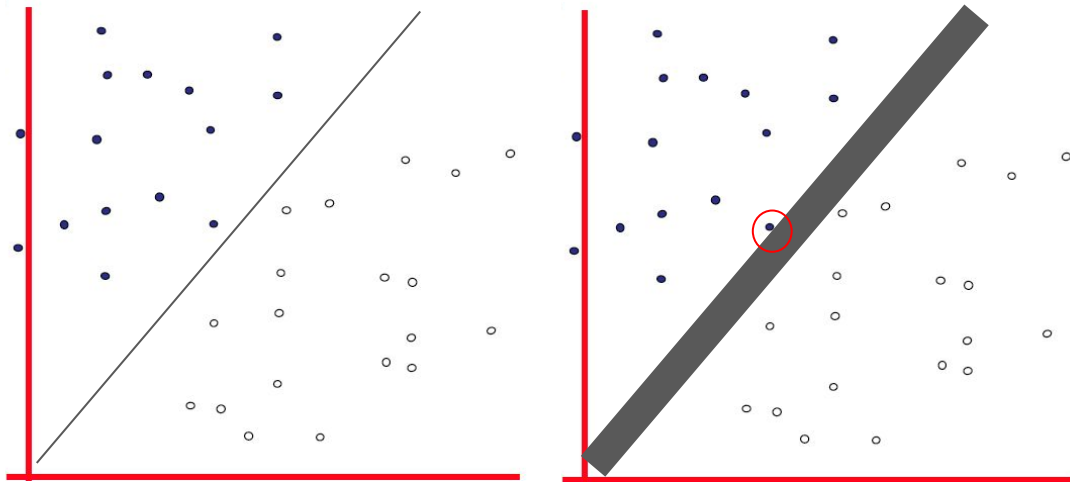
Is there an optimal way to separate the data?

- The theory of support vector machines provides a systematic method for separating the data optimally.
- It involves a key concept called **margin**.



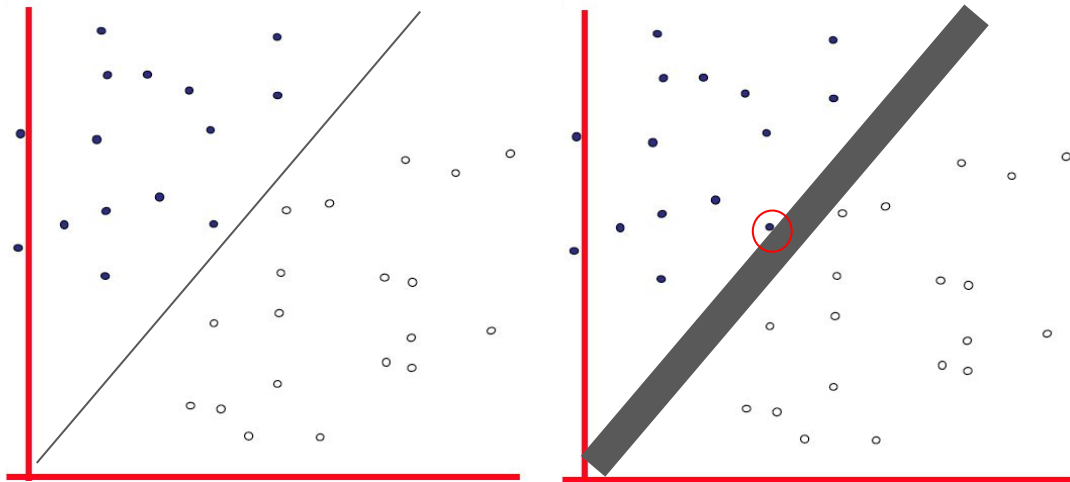
The Margin

- **Margin:** The width that the boundary can be increased before hitting a data point.



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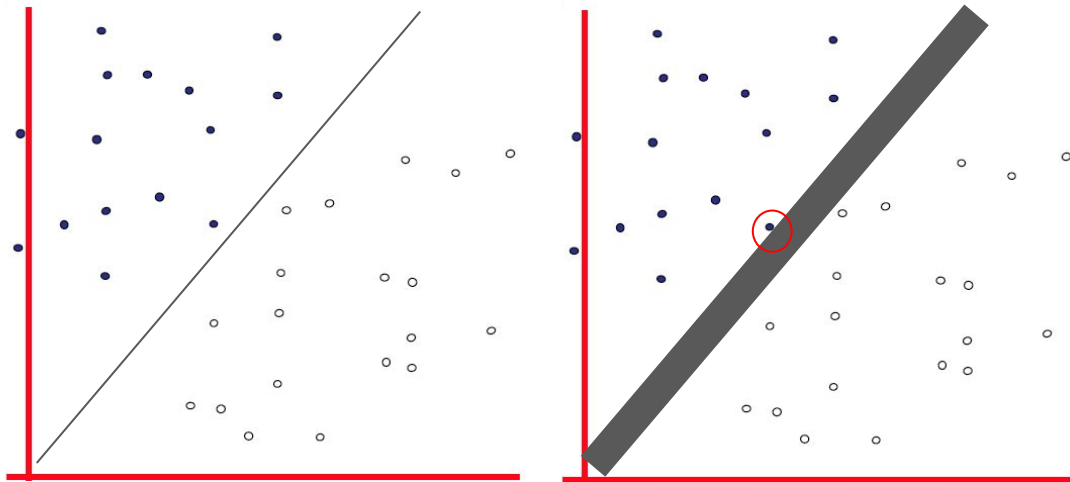


Essentially you are making the hyperplane “fatter” (i.e., increase width) till it hits a data point



The Margin

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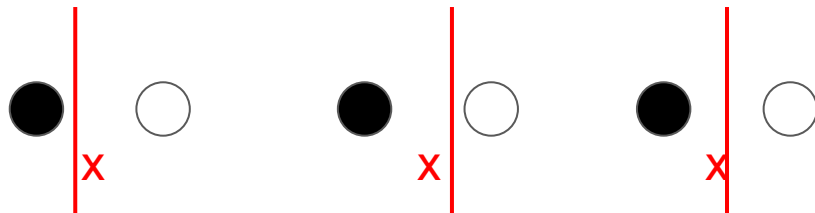


What's the intuition behind this?



The Margin (Intuition)

- A large margin implies that the decision boundary is placed as far away as possible from the data points of each class. This reduces the risk of overfitting to the training data.

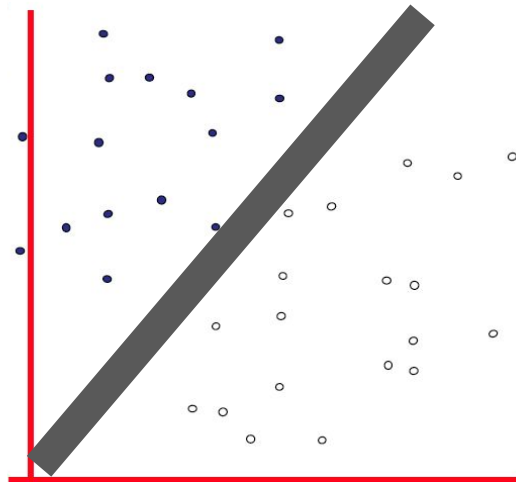


If you have maximum margin, it means the decision boundary is far from the closest point



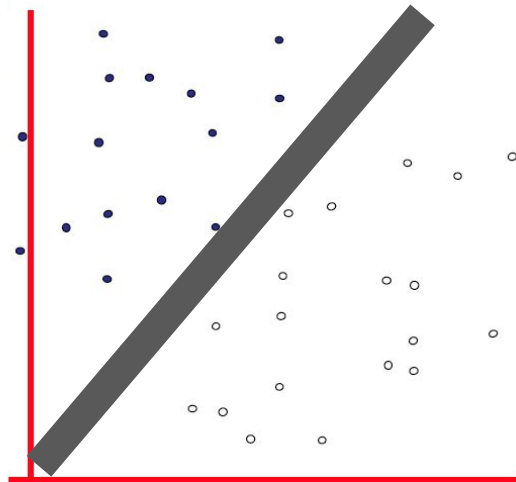
SVM: maximize margin

- The simplest SVM (linear SVM) is the linear classifier with the maximum margin.



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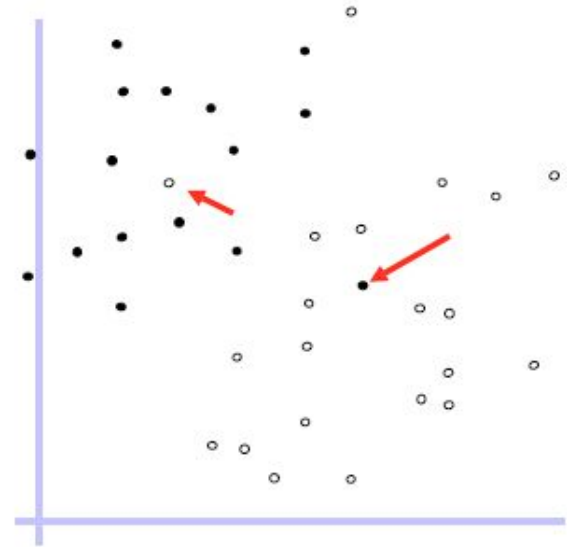
What if the data points are not linearly separable?



SVM: not linearly separable data

Two solutions:

1. Soft margin: Allow a few points on the wrong side (slack variables), and/or
2. Kernel trick: Map data to a higher dimensional space, do linear classification there.



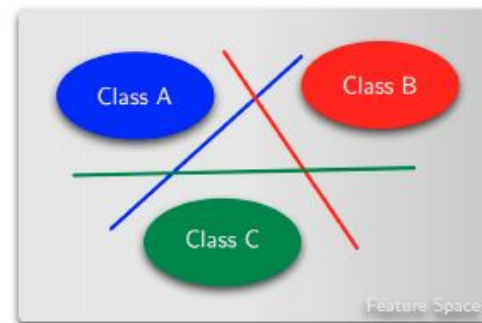
✳ IMPORTANT FOR MID-TERM

SVM: more than two classes

N class problem: Split the task into N binary tasks:

- Class A vs. the rest (class B,...N)
- Class B vs. the rest (class A, B,.. N)
- ...
- Class N vs. the rest

Finally, pick the class that put the point furthest into the positive region.



In this lecture,
we'll work with 2
classes.



SVM Applications

1. Text/document classification
2. Image and video classification
3. Gene Expression Classification
4. Credit Risk Analysis
5. Stock Market Prediction
6. Disease Diagnosis
7. Sentiment classification
8. ... Many more!

SVM: get your hands on it

There are many implementations available:

- <http://svmlight.joachims.org/>
- <https://scikit-learn.org/stable/modules/generated/sklearn.svm.SVC.html>

Technically, you don't have to know the rest of the class in order to use SVM. But...

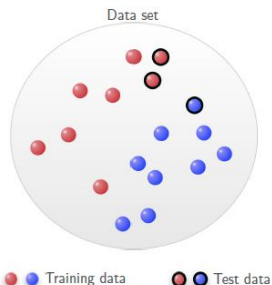


The math version: Support Vector Machines

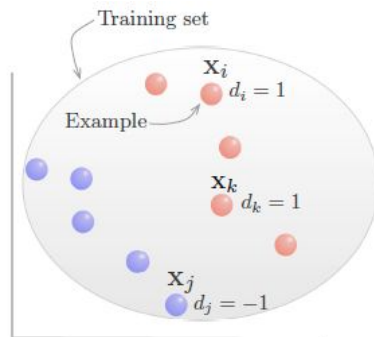
SVM Notation

- Let's assume we have N-dimensional input vector x , where x is equal to $[x_1, \dots, x_N]^T$, with a label $d = \pm 1$.
- SVM parameters: w and $b \longrightarrow \theta$ and θ_0 , in lecture notes.
- The training set can be written as:

$$S = \{(x_1, d_1), (x_2, d_2), \dots, (x_N, d_N)\}$$



Training data is for constructing SVM, and test data is for evaluation.



In this slide, SVM parameters are denoted as w and b . In your lecture notes, they are denoted as θ and θ_0 . It is the same thing!



Classification with a hyperplane $w^T x + b = 0$

A hyperplane, denoted by (w, b) , can be expressed as

$$g(x) = w^T x + b = 0$$

Hyperplane classifies a given x_i with

$$\text{sgn}[g(x_i)] = \begin{cases} +1 & \text{if } g(x_i) > 0 \\ -1 & \text{if } g(x_i) < 0 \end{cases}$$

Hyperplane classifies an example (x_i, d_i) correctly if

substitution:

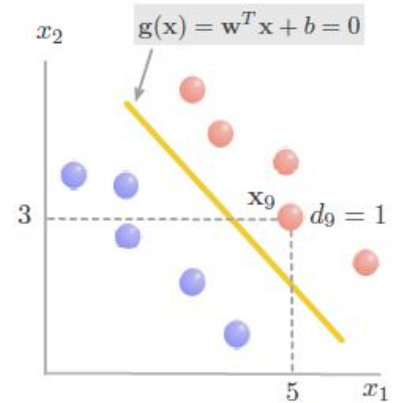
$$\text{sgn}[g(x_i)] = d_i$$

matrix multiplication

Problem: Let $w^T = [7 \ 6]$ and $b = -42$. For $x_9^T = [5 \ 3]$

$$g(x_9) = w^T x_9 + b = [7 \ 6] \begin{bmatrix} 5 \\ 3 \end{bmatrix} - 42 = 11 > 0$$

$$\text{sgn}[g(x_9)] = \text{sgn}[11] = 1 = d_9$$



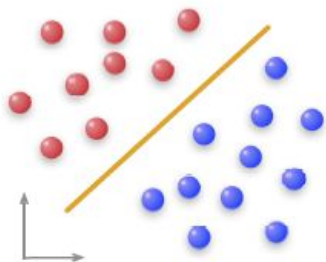
$w^T x + b$
sub w , x & b .
do later!



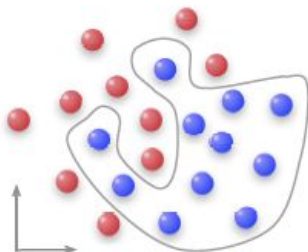
Linearly separable, or not?

Two classes of data are linearly separable if and only if there exists a hyperplane $w^T x + b = 0$ that separates the two classes.

Example in 2D:



Linearly separable



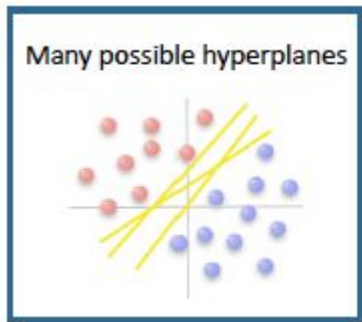
Non-linearly separable

We'll first study the case of "linearly separable".



Margins

The theory of support vector machines provides a systematic method for separating the data optimally, and it involves a key concept called margin.



γ_i^f = Functional margin of an example (x_i, d_i)

γ_i^g = Geometric margin of an example (x_i, d_i)

γ^f = Functional margin of a training set

γ^g = Geometric margin of a training set

γ = Margin of a training set

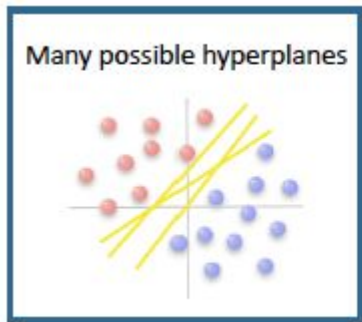
defined w.r.t. a given
hyperplane (w, b)

A training set S can have different geometric margins depending on how hyperplane is defined.

The optimal hyperplane for a given S is the one that gives maximum γ^g over all possible hyperplanes.

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Functional Margin of an example

The functional margin of an example $(\mathbf{x}_i, d_i)$ with respect to a hyperplane $(\mathbf{w}, b)$ is defined as

$$\gamma_i^f = d_i (\mathbf{w}^T \mathbf{x}_i + b)$$

label . sum function

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Problem:

Training set with only 1 example

Hyperplane defined by

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 4 \end{bmatrix}, d_1 = +1$$

$$\mathbf{w} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}, b = 6$$

Determine γ_1^f with respect to given hyperplane

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$$\begin{bmatrix} 5 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix} = (5 \times 1) + (3 \times 4) \\ = 5 + 12 = 17$$

Just need to apply the formulation to perform the calculation! (basic matrix multiplication needed)



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Solution:

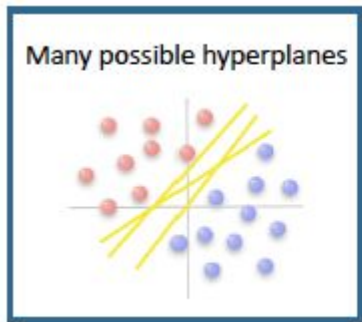
$$\gamma_1^f = d_1 (\mathbf{w}^T \mathbf{x}_1 + b) = 1 \times \left(\begin{bmatrix} 5 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix} + 6 \right) = 1 \times 23 = 23$$

* Substitution : IMPORTANT



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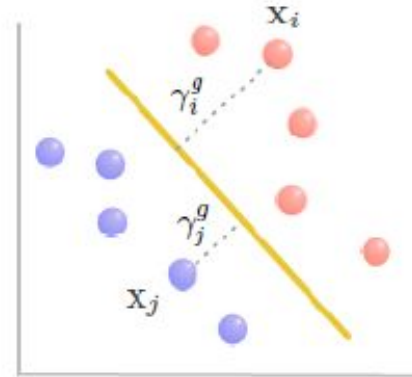
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$$\gamma_i^g = d_i \left(\frac{1}{\|\mathbf{w}\|} \mathbf{w}^T \mathbf{x}_i + \frac{1}{\|\mathbf{w}\|} b \right)$$



γ_i^g is the Euclidean distance from the point defined by $\mathbf{x}_i$ to the hyperplane $(\mathbf{w}, b)$.

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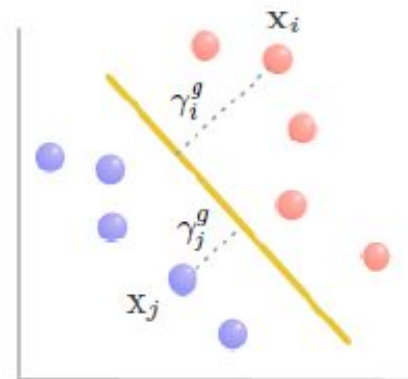
Training set with only 1 example

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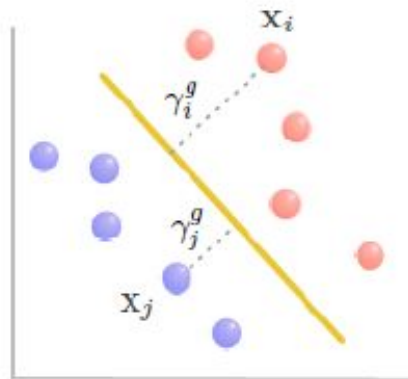
$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 4 \end{bmatrix}, d_1 = +1$$

$$\mathbf{w} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}, b = 6$$

Determine γ_1^g with respect to given hyperplane

Solution:

$$\gamma_1^g = d_1 \left(\frac{\mathbf{w}^T \mathbf{x}_1}{\|\mathbf{w}\|} + \frac{b}{\|\mathbf{w}\|} \right) = \frac{\begin{bmatrix} 5 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix} + 6}{\sqrt{5^2 + 3^2}} = \frac{23}{\sqrt{34}}$$



γ_i^g is the Euclidean distance from the point defined by $\mathbf{x}_i$ to the hyperplane $(\mathbf{w}, b)$.

Just need to apply the formulation to perform the calculation!



*** IMPORTANT**

Functional Margin vs Geometric Margin

Functional margin of an example

$$\gamma_i^f = d_i (\mathbf{w}^T \mathbf{x}_i + b)$$

scalar version
of
 γ_i^f

← Geometric margin of an example

$$\gamma_i^g = d_i \left(\frac{1}{\|\mathbf{w}\|} \mathbf{w}^T \mathbf{x}_i + \frac{1}{\|\mathbf{w}\|} b \right) = \frac{1}{\|\mathbf{w}\|} (\mathbf{w}^T \mathbf{x}_i + b)$$

$$= \frac{1}{\|\mathbf{w}\|} (\gamma_i^f)$$

Therefore, by definition:

$$\gamma_i^g = \frac{\gamma_i^f}{\|\mathbf{w}\|}$$

$$= \frac{\gamma_i^f}{\|\mathbf{w}\|}$$

Scaling Functional & Geometric Margin

Problem: (Note that from earlier slide, $\gamma_1^f = 23$ and $\gamma_1^g = \frac{23}{\sqrt{34}}$)

Training set with only 1 example

Given hyperplane

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 4 \end{bmatrix}, d_1 = +1$$

$$\mathbf{w} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}, b = 6$$

Find scaling constant c such that $\gamma_1^f = 5$ with respect to the given hyperplane, and calculate γ_1^g associated with $c\mathbf{w}$ and cb .

Can we scale
this margin?



Scaling Functional & Geometric Margin

Problem: (Note that from earlier slide, $\gamma_1^f = 23$ and $\gamma_1^g = \frac{23}{\sqrt{34}}$)

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Find scaling constant c such that $\gamma_1^f = 5$ with respect to the given hyperplane, and calculate γ_1^g associated with $c\mathbf{w}$ and cb .

Why?

Solution:

$$d_1 (c \mathbf{w}^T \mathbf{x}_1 + cb) = 1 \left(c \begin{bmatrix} 5 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix} + 6c \right) = 23c = 5 \Rightarrow c = \frac{5}{23}$$

Thus

$$c\mathbf{w} = \frac{5}{23} \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 25/23 \\ 15/23 \end{bmatrix}, cb = \frac{30}{23}$$

and the geometric margin of the example with respect to $c\mathbf{w}$ and cb is

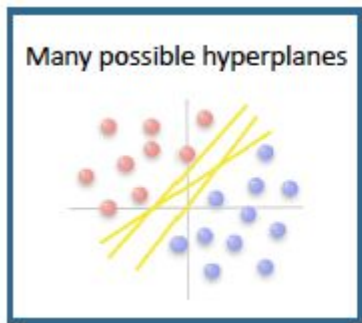
$$\gamma_1^g = \frac{\gamma_1^f}{\|\mathbf{w}\|} = \frac{5}{\sqrt{\left(\frac{25}{23}\right)^2 + \left(\frac{15}{23}\right)^2}} = \frac{5}{\frac{5\sqrt{34}}{23}} = \frac{23}{\sqrt{34}}$$

Can we scale this margin?



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Functional margin of a training set

The functional margin of a training set S , with respect to a hyperplane (w, b) , is the minimum of all the functional margins of the individual examples in S .

$$\gamma_i^f = d_i (w^T x_i + b)$$

$$\gamma^f = \min_{1 \leq i \leq N} \{\gamma_i^f\}$$

$\therefore$ It is the closest point to the decision boundary.

Geometric margin of a training set

The geometric margin of a training set S , with respect to a hyperplane (w, b) , is the minimum of all the geometric margins of the individual examples in S .

$$\gamma_i^g = d_i \left(\frac{1}{\|w\|} w^T x_i + \frac{1}{\|w\|} b \right)$$

$$\gamma^g = \min_{1 \leq i \leq N} \{\gamma_i^g\}$$

Basically, given a hyperplane, you compute for all data points in the training set and find the point with **smallest** margins!



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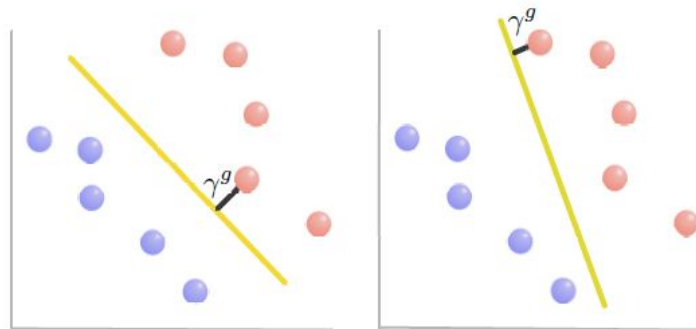
$$\gamma^g = \min_{1 \leq i \leq N} \{\gamma_i^g\}$$

Ehh.. wait.. So many hyperplane, so which is the best one?!



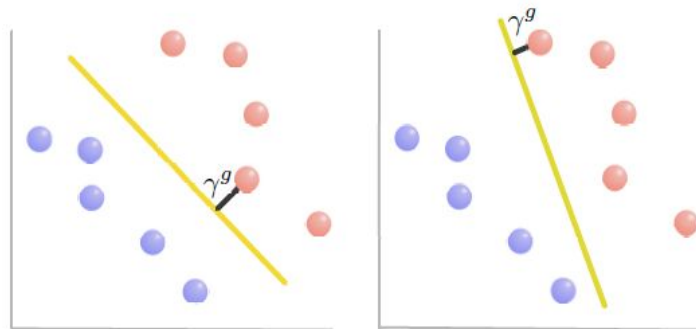
Find the optimal hyperplane

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- The optimal hyperplane for a given S is the one that gives maximum γ^g over all possible hyperplanes.
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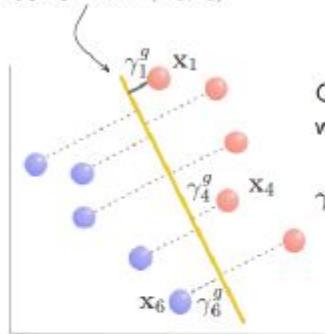


How will you do this in the most naive way?



Step 1: Geometric margin of training set

Hyperplane 1: (w_1, b_1)



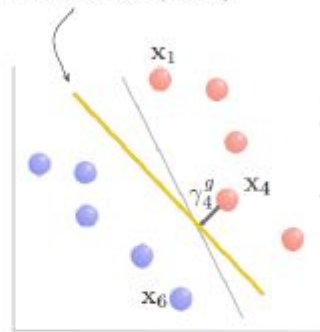
Geometric margin of an example i with respect to hyperplane 1 is

$$\gamma_i^g = d_i \left(\left\langle \frac{1}{\|w_1\|} w_1 \cdot x_i \right\rangle + \frac{1}{\|w_1\|} b_1 \right)$$

Geometric margin of training set with respect to hyperplane 1 is:

$$\min\{\gamma_1^g, \dots, \gamma_{10}^g\} = \gamma_1^g \equiv \gamma_{(w_1, b_1)}^g$$

Hyperplane 2: (w_2, b_2)



Geometric margin of an example i with respect to hyperplane 2 is

$$\gamma_i^g = d_i \left(\left\langle \frac{1}{\|w_2\|} w_2 \cdot x_i \right\rangle + \frac{1}{\|w_2\|} b_2 \right)$$

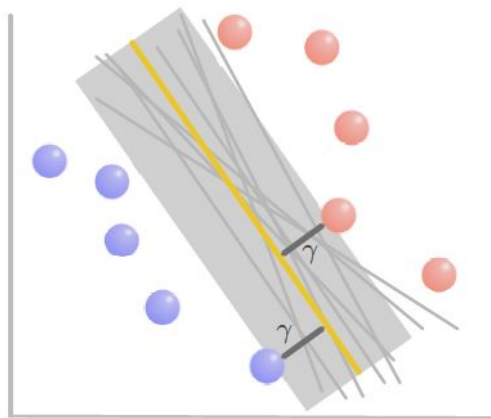
Geometric margin of training set with respect to hyperplane 2 is:

$$\min\{\gamma_1^g, \dots, \gamma_{10}^g\} = \gamma_4^g \equiv \gamma_{(w_2, b_2)}^g$$

Repeat this for all the possible hyperplanes...

Step 2: Margin of training set

- The optimal hyperplane for a given S (data) is one that gives maximum γ^g over all possible hyperplanes



$$\gamma = \max \underbrace{\left\{ \gamma_{(w_1, b_1)}^g, \gamma_{(w_2, b_2)}^g, \gamma_{(w_3, b_3)}^g, \dots \right\}}_{\text{All possible hyperplanes}}$$

Why is it important to find the margin of S? Larger margin leads to lower probability of misclassification.



Finding Margin of S (Mathematically)

- Let's recall the relationship:

$$\gamma_i^g = \frac{\gamma_i^f}{\|w\|}$$

- For a given w , the example k that yields $\gamma_i^f = \gamma^f$ also yields $\gamma_i^g = \gamma^g$. So we can write

$$\uparrow \gamma^g = \frac{\gamma^f}{\|w\|} \downarrow$$

We can maximize γ^g by fixing γ^f then minimizing $\|w\|$

Finding Margin of S (Mathematically)

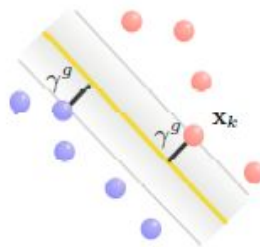
Since a hyperplane is invariant under scaling by a constant c , for **any** hyperplane $(\mathbf{w}, b)$, we can find c such that

$$\gamma^f \equiv \underbrace{\gamma_k^f = d_k (c \mathbf{w}^T \mathbf{x}_k + c b)}_{\text{functional margin of example } k} = 1$$

That is, we fix functional margin γ^f of training set to be 1.

Problem: Find c such that $\gamma_k^f = 1$, with

$$\mathbf{x}_k = \begin{bmatrix} 1 \\ 4 \end{bmatrix}, \quad d_k = +1, \quad \mathbf{w} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}, \quad b = 6$$



$$d_k (c \mathbf{w}^T \mathbf{x}_k + c b) = 1 \cdot c \left(\begin{bmatrix} 5 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix} + 6 \right) = 23c = 1 \Rightarrow c = \frac{1}{23}$$

We are doing the scaling for easy computation



Finding Margin of S (Mathematically)

- With γ^f , fixed at 1, for any example x_i , we have the following condition

$$\gamma_i^f = \gamma_i^g \|w\| = d_i (w^T x_i + b) \geq 1$$

Recall

$$\gamma_i^f = d_i (w^T x_i + b)$$

- The problem of minimizing $\|w\|$ can now be expressed as a

Constrained optimization problem

$$\text{Minimizing: } f(w) = \frac{1}{2} \|w\|^2 = \frac{1}{2} w^T w$$

$$\text{Subject to: } d_i (w^T x_i + b) \geq 1$$

$$\gamma^f = \min_{1 \leq i \leq N} \{ \gamma_i^f \}$$

- By solving this problem, we can obtain the optimal hyperplane (w_0, b_0)

Discriminant function and support vector

- After obtaining w_0 , b_0 , for training data x_i , classification will be as follows:

$$\begin{aligned} g(x_i) &= (w_0^T x_i + b_0) \geq 1 \text{ for } d_i = +1 \\ g(x_i) &= (w_0^T x_i + b_0) < -1 \text{ for } d_i = -1 \end{aligned}$$

- Discrimin

- Support vector: x_i th

$$g(x) = w_0^T x + b_0$$

$$g(x_i) = \pm 1$$

Minimizing $\|\mathbf{w}\|$: Constrained optimization

Primal problem

Given data set : $S = \{(\mathbf{x}_i, d_i)\}, i = 1, 2, \dots, N$

Find : $\mathbf{w}$ and b

Minimizing : $f(\mathbf{w}) = \frac{1}{2}\mathbf{w}^T\mathbf{w}$

Subject to : $d_i (\mathbf{w}^T \mathbf{x}_i + b) \geq 1$

We will solve this problem with Lagrange Theorem. Solving this problem yields the optimal hyperplane $(\mathbf{w}_o, b_o)$!



Joseph-Louis Lagrange
French mathematician
1736–1813

Lost? Take a break to find
back your soul before we
continue the adventure!

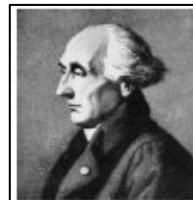


Minimizing $\|w\|$: Constrained optimization

Lagrange Theorem:

Minimize: $f(w)$

Subject to: $h_i(w) = 0, i = 1, 2, \dots, m$



Joseph-Louis Lagrange
French mathematician
1736–1813

Let's define the **Lagrangian Function**:

$$L(w, \beta) = f(w) + \sum_{i=1}^m \beta_i h_i(w)$$

The constant β_i are called **Lagrange multipliers**

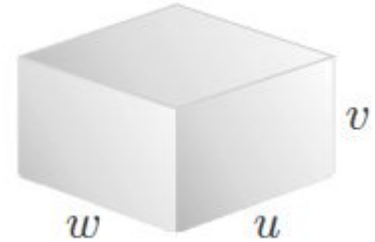
The necessary conditions for the existence of an optimal solution w_0 corresponding to a minimum of $f(w)$ subject to $h_i(w) = 0, i = 1, 2, \dots, m$ are:

$$\frac{\partial L(w_0, \beta_0)}{\partial w} = 0$$

$$\frac{\partial L(w_0, \beta_0)}{\partial \beta_i} = 0$$

A simple example

Consider a box whose surface area is C . Formulate the optimization problem for maximizing the volume of the box.



A simple example

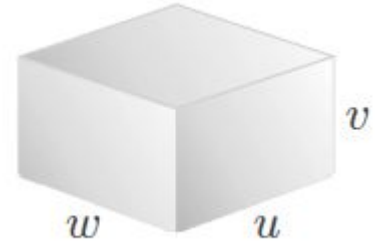
Consider a box whose surface area is C . Formulate the optimization problem for maximizing the volume of the box.

Solution:

Surface area: $2wu + 2uv + 2vw = C$

Volume: wuv

Maximizing wuv is equivalent to minimizing $-wuv$.



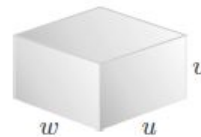
Optimization problem

$$\begin{aligned} \text{Minimize: } & f(u, v, w) = -uvw \\ \text{Subject to: } & h(u, v, w) = wu + uv + vw - \frac{C}{2} = 0 \end{aligned}$$

A simple example

Minimize: $f(u, v, w) = -uvw$

Subject to: $h(u, v, w) = wu + uv + vw - \frac{C}{2} = 0$

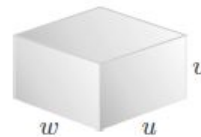


$$L(u, v, w) = -uvw + \beta \left(wu + uv + vw - \frac{C}{2} \right)$$

A simple example

Minimize: $f(u, v, w) = -uvw$

Subject to: $h(u, v, w) = wu + uv + vw - \frac{C}{2} = 0$



$$L(u, v, w) = -uvw + \beta \left(wu + uv + vw - \frac{C}{2} \right)$$

The solution is

Lagrange's Theorem yields

$$\frac{\partial L}{\partial u} = -vw + \beta(v + w) = 0$$

$$\frac{\partial L}{\partial v} = -wu + \beta(w + u) = 0$$

$$\frac{\partial L}{\partial w} = -uv + \beta(u + v) = 0$$

$$wu + uv + vw = \frac{C}{2}$$

$$u = v = w = \sqrt{\frac{C}{6}}$$

and the maximum volume is

$$\begin{aligned} \max(wuv) &= -\min(-wuv) \\ &= -\left(-\left(\frac{C}{6}\right)^{\frac{3}{2}}\right) = \left(\frac{C}{6}\right)^{\frac{3}{2}} \end{aligned}$$

Generalized Lagrangian Function & KKT conditions

Lagrange Theorem can be generalized to deal with problems having both equality and inequality constraints.

$$\begin{array}{ll}\text{Minimize:} & f(\mathbf{w}) \\ \text{Subject to:} & q_i(\mathbf{w}) \leq 0, \quad i = 1, \dots, k \\ & h_i(\mathbf{w}) = 0, \quad i = 1, \dots, m\end{array}$$

Generalized Lagrangian function

$$L(\mathbf{w}, \alpha, \beta) = f(\mathbf{w}) + \sum_{i=1}^k \alpha_i q_i(\mathbf{w}) + \sum_{i=1}^m \beta_i h_i(\mathbf{w})$$

The constants α_i, β_i are called Lagrange multipliers. The solution is characterized by **Karush-Kuhn-Tucker (KKT)** conditions.

Generalized Lagrangian Function & KKT conditions

Kuhn-Tucker Theorem

Consider the primal problem with the Lagrangian as defined earlier. The sufficient and necessary condition for a point $\mathbf{w}_o$ to be an optimal solution is the existence of α_o and β_o such that

$$\frac{\partial L(\mathbf{w}_o, \alpha_o, \beta_o)}{\partial \mathbf{w}} = 0$$

$$\frac{\partial L(\mathbf{w}_o, \alpha_o, \beta_o)}{\partial \beta_i} = 0$$

$$\alpha_{o,i} q_i(\mathbf{w}_o) = 0, \quad i = 1, \dots, k$$

$$q_i(\mathbf{w}_o) \leq 0, \quad i = 1, \dots, k$$

$$\alpha_{o,i} \geq 0, \quad i = 1, \dots, k$$

WHY KKT? Because this will let us solve the problem by computing the just the inner products of x_i, x_j (which will be very important later on when we want to solve non-linearly separable classification problems)



Primal problem vs dual problem

The KKT conditions enable us to transform the primal problem into an alternative, simpler form called the **dual problem**.

Primal problem

Minimize: $f(\mathbf{w})$

Subject to: $q_i(\mathbf{w}) \leq 0$

$h_i(\mathbf{w}) = 0$

Dual problem

Maximize: $L(\mathbf{w}, \alpha, \beta)$

Subject to: $\frac{\partial L(\mathbf{w}_o, \alpha_o, \beta_o)}{\partial \mathbf{w}} = 0$

$\alpha \geq 0$

The primal and dual problems are equivalent because they have the same value of the optimization problem:

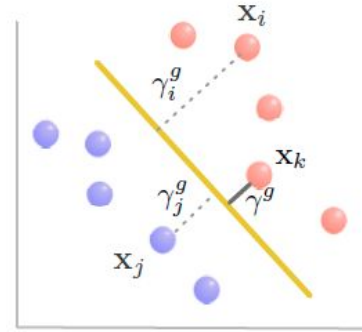


$$f(\mathbf{w}_o) = L(\mathbf{w}_o, \alpha_o, \beta_o)$$

The KKT conditions enable us to transform the primal problem into an alternative, simpler form called the dual problem

Let's recall our primal problem:

Given data set : $S = \{(x_i, d_i)\}$
Find : w and b
Minimizing : $f(w) = \frac{1}{2}w^T w$
Subject to : $d_i (w^T x_i + b) \geq 1$



In primal problem, known parameters are x_i , d_i and unknown variables are w , b .

Apply KKT conditions to reduce unknowns to just α_i , and to form a **dual problem**.



Transforming primal problem into dual problem by KKT conditions

Step 1: Write constraint $d_i(w^T x_i + b) \geq 1$ in a standard form as:

$$-d_i(w^T x_i + b) + 1 \leq 0$$

Transforming primal problem into dual problem by KKT conditions

Step 1: Write constraint $d_i(w^T x_i + b) \geq 1$ in a standard form as:

$$-d_i(w^T x_i + b) + 1 \leq 0$$

Step 2: Write the Lagrangian function

$$\begin{aligned} L(w, b, \alpha) &= \frac{1}{2} w^T w - \sum_{i=1}^N \alpha_i (d_i (w^T x_i + b) - 1) \\ &= \frac{1}{2} w^T w - \sum_{i=1}^N \alpha_i d_i w^T x_i - b \sum_{i=1}^N \alpha_i d_i + \sum_{i=1}^N \alpha_i \end{aligned}$$

Transforming primal problem into dual problem by KKT conditions

Step 1: Write constraint $d_i(w^T x_i + b) \geq 1$ in a standard form as:

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Step 3: KKT conditions are:

$$\frac{\partial L(w, b, \alpha)}{\partial w} = 0$$

$$\frac{\partial L(w, b, \alpha)}{\partial b} = 0$$

$$d_i (w^T x_i + b) \geq 1$$

$$\alpha_i (d_i (w^T x_i + b) - 1) = 0$$

$$\alpha_i \geq 0$$

Transforming primal problem into dual problem by KKT conditions

Step 3: KKT conditions are:

$$\frac{\partial L(\mathbf{w}, b, \alpha)}{\partial \mathbf{w}} = 0$$

$$\frac{\partial L(\mathbf{w}, b, \alpha)}{\partial b} = 0$$

$$d_i (\mathbf{w}^T \mathbf{x}_i + b) \geq 1$$

$$\alpha_i (d_i (\mathbf{w}^T \mathbf{x}_i + b) - 1) = 0$$

$$\alpha_i \geq 0$$

Let's focus on
each condition
one by one



Transforming primal problem into dual problem by KKT conditions

The first condition:

$$\begin{aligned}\frac{\partial L(\mathbf{w}, b, \alpha)}{\partial \mathbf{w}} &= \frac{\partial}{\partial \mathbf{w}} \left(\frac{\mathbf{w}^T \mathbf{w}}{2} - \sum_{i=1}^N \alpha_i d_i \mathbf{w}^T \mathbf{x}_i - b \sum_{i=1}^N \alpha_i d_i + \sum_{i=1}^N \alpha_i \right) \\ &= \frac{1}{2} \frac{\partial (\mathbf{w}^T \mathbf{w})}{\partial \mathbf{w}} - \sum_{i=1}^N \alpha_i d_i \left(\frac{\partial (\mathbf{w}^T \mathbf{x}_i)}{\partial \mathbf{w}} \right) \\ &= \frac{2\mathbf{w}}{2} - \sum_{i=1}^N \alpha_i d_i \left(\frac{\partial (\mathbf{x}_i^T \mathbf{w})}{\partial \mathbf{w}} \right) \quad (\text{Since } \mathbf{u}^T \mathbf{v} = \mathbf{v}^T \mathbf{u}) \\ &= \mathbf{w} - \sum_{i=1}^N \alpha_i d_i \mathbf{x}_i \quad \left(\text{Since } \frac{\partial (\mathbf{u}^T \mathbf{v})}{\partial \mathbf{v}} = \mathbf{u} \right) \\ &= \mathbf{0}\end{aligned}$$

Therefore:

$$\mathbf{w} = \sum_{i=1}^N \alpha_i d_i \mathbf{x}_i$$

Transforming primal problem into dual problem by KKT conditions

Step 3: KKT conditions are:

$$\frac{\partial L(\mathbf{w}, b, \alpha)}{\partial \mathbf{w}} = 0$$

$$\frac{\partial L(\mathbf{w}, b, \alpha)}{\partial b} = 0$$

$$d_i (\mathbf{w}^T \mathbf{x}_i + b) \geq 1$$

$$\alpha_i (d_i (\mathbf{w}^T \mathbf{x}_i + b) - 1) = 0$$

$$\alpha_i \geq 0$$

Transforming primal problem into dual problem by KKT conditions

The second condition:

$$\begin{aligned} & \frac{\partial L(\mathbf{w}, b, \alpha)}{\partial b} \\ &= \frac{\partial}{\partial b} \left(\frac{1}{2} \mathbf{w}^T \mathbf{w} - \sum_{i=1}^N \alpha_i d_i \mathbf{w}^T \mathbf{x}_i - b \sum_{i=1}^N \alpha_i d_i + \sum_{i=1}^N \alpha_i \right) \\ &= \sum_{i=1}^N \alpha_i d_i = 0 \end{aligned}$$

Therefore:

$$\sum_{i=1}^N \alpha_i d_i = 0$$

Transforming primal problem into dual problem by KKT conditions

Let's recall the Lagrangian function

$$L(\mathbf{w}, b, \alpha) = \frac{1}{2} \mathbf{w}^T \mathbf{w} - \sum_{i=1}^N \alpha_i d_i \mathbf{w}^T \mathbf{x}_i - b \sum_{i=1}^N \alpha_i d_i + \sum_{i=1}^N \alpha_i$$

And remember:

$$\mathbf{w} = \sum_{i=1}^N \alpha_i d_i \mathbf{x}_i$$

$$\sum_{i=1}^N \alpha_i d_i = 0$$

So:

$$\frac{1}{2} \mathbf{w}^T \mathbf{w} = \frac{1}{2} \left[\sum_{i=1}^N \alpha_i d_i \mathbf{x}_i^T \right] \left[\sum_{j=1}^N \alpha_j d_j \mathbf{x}_j \right] = \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j d_i d_j \mathbf{x}_i^T \mathbf{x}_j$$

$$\sum_{i=1}^N \alpha_i d_i \mathbf{w}^T \mathbf{x}_i = \sum_{i=1}^N \alpha_i d_i \left[\sum_{j=1}^N \alpha_j d_j \mathbf{x}_j^T \right] \mathbf{x}_i = \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j d_i d_j \mathbf{x}_i^T \mathbf{x}_j$$

Transforming primal problem into dual problem by KKT conditions

Let's recall the Lagrangian function

$$L(\mathbf{w}, b, \alpha) = \frac{1}{2} \mathbf{w}^T \mathbf{w} - \sum_{i=1}^N \alpha_i d_i \mathbf{w}^T \mathbf{x}_i - b \sum_{i=1}^N \alpha_i d_i + \sum_{i=1}^N \alpha_i$$

And remember:

$$\mathbf{w} = \sum_{i=1}^N \alpha_i d_i \mathbf{x}_i$$

$$\sum_{i=1}^N \alpha_i d_i = 0$$

So:

$$\begin{aligned} \frac{1}{2} \mathbf{w}^T \mathbf{w} &= \frac{1}{2} \left[\sum_{i=1}^N \alpha_i d_i \mathbf{x}_i^T \right] \left[\sum_{j=1}^N \alpha_j d_j \mathbf{x}_j \right] = \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j d_i d_j \mathbf{x}_i^T \mathbf{x}_j \\ \sum_{i=1}^N \alpha_i d_i \mathbf{w}^T \mathbf{x}_i &= \sum_{i=1}^N \alpha_i d_i \left[\sum_{j=1}^N \alpha_j d_j \mathbf{x}_j^T \right] \mathbf{x}_i = \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j d_i d_j \mathbf{x}_i^T \mathbf{x}_j \end{aligned}$$

Don't faint yet!
It's not so chim!
We are just
substituting the
terms!



Transforming primal problem into dual problem by KKT conditions

The Lagrangian function can be written as:

$$\begin{aligned} L(\mathbf{w}, b, \alpha) &= \frac{1}{2} \mathbf{w}^T \mathbf{w} - \sum_{i=1}^N \alpha_i (d_i (\mathbf{w}^T \mathbf{x}_i + b) - 1) \\ &= \frac{1}{2} \mathbf{w}^T \mathbf{w} - \sum_{i=1}^N \alpha_i d_i \mathbf{w}^T \mathbf{x}_i - b \sum_{i=1}^N \alpha_i d_i + \sum_{i=1}^N \alpha_i \\ &= \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j d_i d_j \mathbf{x}_i^T \mathbf{x}_j \\ &\equiv Q(\alpha) \end{aligned}$$

Given data set : $S = \{(\mathbf{x}_i, d_i)\}$

Find : $\mathbf{w}$ and b

Minimizing : $f(\mathbf{w}) = \frac{1}{2} \mathbf{w}^T \mathbf{w}$

Subject to : $d_i (\mathbf{w}^T \mathbf{x}_i + b) \geq 1$

Finding optimal hyperplane (dual problem)

Given : $S = \{(\mathbf{x}_i, d_i)\}$

Find : Lagrange multipliers $\{\alpha_i\}$

Maximizing : $Q(\alpha) = \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j d_i d_j \mathbf{x}_i^T \mathbf{x}_j$

Subject to : (1) $\sum_{i=1}^N \alpha_i d_i = 0$
(2) $\alpha_i \geq 0$

1. Algorithms and software available for solving this problem.
2. α_i are the only unknowns.
3. $\mathbf{x}_i^T \mathbf{x}_j$ is called a linear kernel

An Example

Suppose that a support vector machine is to be constructed using the training set Below:

- a) Determine the Lagrangian function
- b) Show the explicit form of the dual problem

i	$\mathbf{x}_i$	d_i
1	$[1 \ -1]^T$	-1
2	$[2 \ 1]^T$	1
3	$[3 \ 1]^T$	1



An Example

Let's remember the general formulation of Lagrangian function.

$$L(\mathbf{w}, b, \alpha) = \frac{1}{2} \mathbf{w}^T \mathbf{w} - \sum_{i=1}^N \alpha_i (d_i (\mathbf{w}^T \mathbf{x}_i + b) - 1)$$

From the given data, we have $N = 3$,
and $\mathbf{w} = [w_1, w_2]^T$

$$\begin{aligned} L(\mathbf{w}, b, \alpha) &= \frac{1}{2} \begin{bmatrix} w_1 & w_2 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \\ &\quad - \alpha_1(-1) \begin{bmatrix} w_1 & w_2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} - \alpha_1(-1)b + \alpha_1 \\ &\quad - \alpha_2(1) \begin{bmatrix} w_1 & w_2 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} - \alpha_2(1)b + \alpha_2 \\ &\quad - \alpha_3(1) \begin{bmatrix} w_1 & w_2 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} - \alpha_3(1)b + \alpha_3 \\ &= \frac{1}{2} (w_1^2 + w_2^2) + \alpha_1 (w_1 - w_2 + 1) - \alpha_2 (2w_1 + w_2 - 1) \\ &\quad - \alpha_3 (3w_1 + w_2 - 1) + (\alpha_1 - \alpha_2 - \alpha_3) b \end{aligned}$$

An Example

Explicit form of Dual problem

Maximize

$$Q(\alpha) = \alpha_1 + \alpha_2 + \alpha_3 - \frac{1}{2} (2\alpha_1^2 + 5\alpha_2^2 + 10\alpha_3^2 - 2\alpha_1\alpha_2 - 4\alpha_1\alpha_3 + 14\alpha_2\alpha_3)$$

Subject to

$$\sum_{i=1}^3 \alpha_i d_i = \alpha_1(-1) + \alpha_2(1) + \alpha_3(1) = -\alpha_1 + \alpha_2 + \alpha_3 = 0$$

$$\alpha_1 \geq 0$$

$$\alpha_2 \geq 0$$

$$\alpha_3 \geq 0$$

Finding optimal hyperplane (dual problem)

Given : $S = \{(\mathbf{x}_i, d_i)\}$

Find : Lagrange multipliers $\{\alpha_i\}$

Maximizing : $Q(\alpha) = \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j d_i d_j \mathbf{x}_i^T \mathbf{x}_j$

Subject to : (1) $\sum_{i=1}^N \alpha_i d_i = 0$

(2) $\alpha_i \geq 0$

Lagrange multiplier and support vector

Support vectors are the data points that lie closest to the decision

Recall:

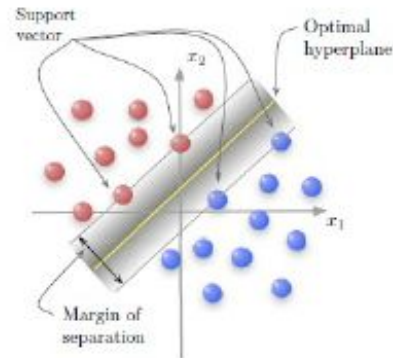
For support vector x_i : $g(x_i) = w_0^T x_i + b = \pm 1$

From the 3rd KKT condition: $d_i(w_0^T x_i + b_0) \geq 1$

From the 4th KKT condition: $\alpha_{0,i} (d_i(w_0^T x_i + b_0) - 1) = 0$

For data point x_i that is a support vector:

$$d_i(w_0^T x_i + b_0) = 1 \text{ AND } \alpha_{0,i} \neq 0$$



Most of the α_i values will turn out to have the value zero. The non-zero α_i will correspond to the support vectors. For a support vector x_i :

$$d_i(w_0^T x_i + b_0) = 1 \quad \longrightarrow \quad b_0 = \frac{1}{d_i} - w_0^T x_i$$

Optimal solution

Primal	Dual
Find : $\mathbf{w}, b$	α_i
Minimizing : $f(\mathbf{w})$	$\mathbf{w}$
Maximizing : $-$	$Q(\alpha)$
Subject to : $d_i (\mathbf{w}^T \mathbf{x}_i + b) \geq 1$	$\sum_{i=1}^N \alpha_i d_i = 0$
	$\alpha_i \geq 0$

$$f(\mathbf{w}, \xi) = \frac{1}{2} \mathbf{w}^T \mathbf{w} \quad Q(\alpha) = \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j d_i d_j \mathbf{x}_i^T \mathbf{x}_j$$

Dual Problem formulation

After $\alpha_{o,i}$ is obtained, we can calculate $\mathbf{w}_o$ and b_o as follows:

$$\mathbf{w}_o = \sum_{i=1}^N \alpha_{o,i} d_i \mathbf{x}_i, \quad b_o = \frac{1}{d^{(s)}} - \mathbf{w}_o^T \mathbf{x}^{(s)}$$

where $\mathbf{x}^{(s)}$ is a support vector with label $d^{(s)}$

That's all folks! We live,
and continue the
adventure another day!



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 - *University of Edinburgh, United Kingdom – Machine Learning And Pattern Recognition (MLPR, INFR11130)*
 - *Stanford University – Machine Learning (CS229)*